

Feature Engineering

Project: Responsible Plant Classifier

1. Your mission

Your learning target

I can create features available at prediction time.

I can fit feature transformations using training evidence only.

I can compare a feature change without changing the evaluation set.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Feature engineering transforms raw inputs into useful structure while preserving prediction-time availability and avoiding leakage.



LEARNING TARGET Length=12 cm, width=30 mm. Compute length/width using matching units. Explain handling for zero or missing width and where learned scaling is fitted.

Builder name and experiment date:

Readiness for the connected explanation

Why must preprocessing be fitted on training only?

Readiness for Relationships and confounding

Can a survey of only one club establish the preference of all students?

My readiness explanation

This week's end goal

By the end, I can explain feature engineering using a traced example and a stated limitation.

2. Notice the evidence

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible evidence; do not explain more than the picture and information support.



MYSTERY The team must use feature engineering to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

A closer explanation

A useful feature represents information available when the prediction is made. In a fictional parcel task, length and width are measured before packing, so $\text{area} = \text{length} \times \text{width}$ is available. The final delivery delay is not available when predicting dispatch time. A strong correlation cannot make a future value legitimate.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines feature engineering.

Apply the connected idea: Relationships and confounding

In a fictional log, groups that practiced longer also scored higher: 1 hour with 4 points, 2 hours with 6 and 3 hours with 8. This pattern is an association. Prior experience could affect both practice choices and scores.

The table alone cannot establish the effect of adding practice time.

3. See how it works

ML MECHANISM

Input → **Model** → **Evidence**

Trace the information, fitted behavior, and evaluation proof separately.



IN

INPUT

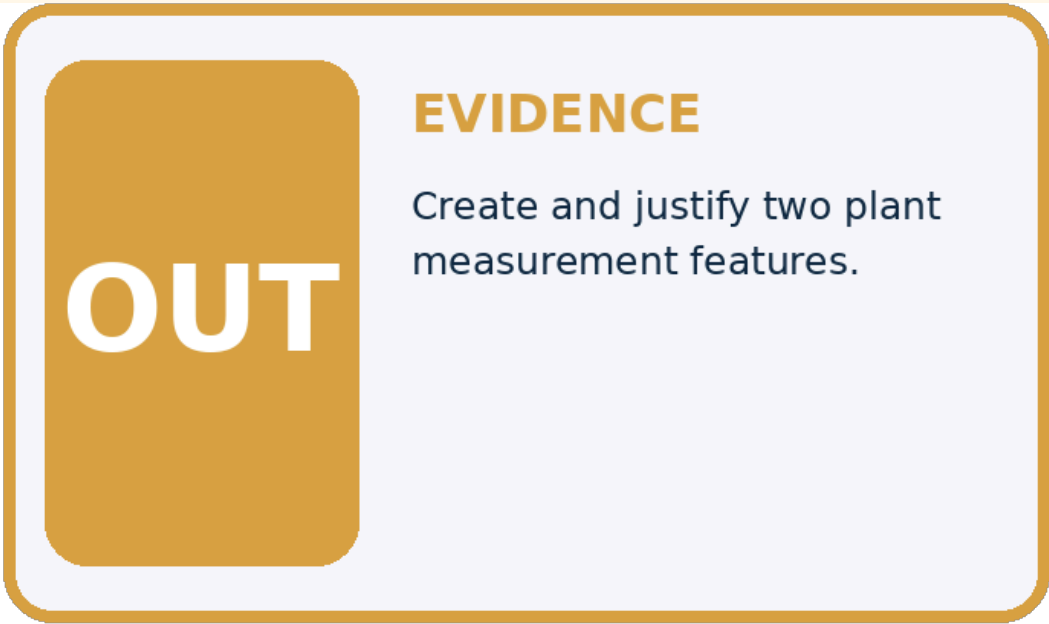
Create and justify two plant measurement features.



ML

MODEL

Feature engineering transforms raw inputs into useful structure while preserving prediction-time availability and avoiding leakage.



OUT

EVIDENCE

Create and justify two plant measurement features.

QUESTION Create and justify two plant measurement features.

MODEL IDEA Feature engineering transforms raw inputs into useful structure while preserving prediction-time availability and avoiding leakage.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Raw leaf length=8 cm and width=4 cm. Derived aspect_ratio = $4 \div 8 = 0.5$.

Draw a fresh data → model → prediction → evaluation trace:

A closer explanation

Training rectangles have lengths [2, 4, 6] and widths [3, 2, 1], so areas are [6, 8, 6] square cm. Training mean area is $20/3$. A validation rectangle length 5 width 2 has area 10 and centered area $10 - 20/3 = 10/3$. Freeze the training mean; do not recompute it with validation.

Before continuing, underline the decision rule and circle the evidence that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim. Keep the distinction between an observed result and a proposed next step.

Apply the connected idea: Relationships and confounding

```
rows = [{"hours": 1, "score": 4},
        {"hours": 2, "score": 6},
        {"hours": 3, "score": 8}]

for row in rows:
    print(row["hours"], row["score"])
```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: Observed

Evidence: More practice with higher scores

Meaning: Association

Step or case: Possible other factor

Evidence: Prior experience

Meaning: May affect both

Step or case: Missing evidence

Evidence: Comparable starting conditions

Meaning: Limits causal claim

Read each row from left to right and connect the evidence to the stated rule.

The numbers rise together in these three invented records. A description such as higher practice hours went with higher scores is supported. A claim that one extra hour guarantees two more points goes beyond the observations. No statistical calculation is required to see this limit.

Separate a pattern from a cause

The three records show that higher practice time goes with higher scores. They do not tell us what would happen if the same learner practised for one more hour. Prior experience could make someone both practise more and score higher. That is a possible confounder: another influence connected to both quantities. First describe the pattern accurately. Then propose a comparison that addresses alternative explanations rather than treating the pattern as a guarantee.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

EVALUATION DESIGN

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL CASE

Expected data from the intended setting



BOUNDARY CASE

Value near a threshold or class boundary



MISSING CASE

Unknown field, label, or necessary evidence



SHIFTED CASE

New subgroup, device, condition, or time

NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change: Include the query in the centering mean.

Prediction: Mean becomes $26/4 = 6.5 \text{ cm}^2$ and query feature 1.5 cm^2 ; this changes the fitted transformation.

Reason: A held-out input should not silently redefine a training-fitted preprocessing step.

5. Words that unlock the idea

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.



2

DERIVED FEATURE

new input calculated from
existing inputs

3

ASPECT RATIO

width divided by length

4

FEATURE PIPELINE

fixed ordered
transformations

Term · Precise meaning · My ML example

Term: transformation

Precise meaning: rule converting raw values

My ML example: _____

Term: derived feature

Precise meaning: new input calculated from existing inputs

My ML example: _____

Term: aspect ratio

Precise meaning: width divided by length

My ML example: _____

Term: feature pipeline

Precise meaning: fixed ordered transformations

My ML example: _____

MISCONCEPTION TO REPAIR Creating many features automatically gives the model more useful information.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word · Meaning

Word: Feature engineering

Meaning: Constructing model inputs from available measurements

Word: Derived feature

Meaning: An input computed from other inputs

Word: Leakage

Meaning: Information entering a model that would not be available legitimately

Word: Availability

Meaning: Whether a value exists when a prediction is needed

Terms for Relationships and confounding

Term · Meaning

Term: Association

Meaning: Two variables vary together in observed data

Term: Cause

Meaning: A change in one factor produces a change in another

Term: Confounder

Meaning: A factor related to both the proposed cause and outcome

Term: Comparison

Meaning: Examining outcomes under different conditions

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

WEEK 29 • WORKED EVIDENCE

- 1 Raw leaf length=8 cm and width=4 cm.
- 2 Derived aspect_ratio = $4 \div 8 = 0.5$.
- 3 Color histogram summarizes pixel ranges without using the target label.
- 4 Fit transformation settings on training data only; document unit, purpose, availability, and leakage check.

STEP 1 Raw leaf length=8 cm and width=4 cm.

STEP 2 Derived aspect_ratio = $4 \div 8 = 0.5$.

STEP 3 Color histogram summarizes pixel ranges without using the target label.

STEP 4 Fit transformation settings on training data only; document unit, purpose, availability, and leakage check.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

A closer explanation

Feature engineering at a glance

Part · Evidence or action · Interpretation

Part: Raw measurements

Evidence or action: Length and width

Interpretation: Available before prediction

Part: Derived area

Evidence or action: Multiply length by width

Interpretation: Units square cm

Part: Centering

Evidence or action: Subtract training mean

Interpretation: Reuse frozen value

Area is a deterministic transformation; it adds a representation, not a new measured fact. It may help a limited model express a relationship, or may do nothing useful. Compare original versus extended features under the same split, metric and development policy.

Center the validation area using the frozen training mean. Explain the steps or evidence behind your response.

Apply the connected idea: Relationships and confounding

Name one factor besides practice duration that could influence both time chosen and score.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Feature Engineering

Engineer a feature with a transformation defined before evaluating a new leaf.

The facts we will use

Training rectangular leaf cards have areas [2, 6, 10] cm². Area=length×width; center by training mean. Query length 2, width 4 cm.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Feature Engineering

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Construct area

Query area = $2 \times 4 = 8 \text{ cm}^2$.



2. Fit centering

Training mean = $(2+6+10)/3 = 6 \text{ cm}^2$.



3. Apply unchanged transform

Centered query feature = $8 - 6 = 2 \text{ cm}^2$.

CHANGE ONE THING

Include the query in the centering mean.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Construct area

What happens and why: Query area = $2 \times 4 = 8 \text{ cm}^2$.

Step: 2. Fit centering

What happens and why: Training mean = $(2+6+10)/3 = 6 \text{ cm}^2$.

Step: 3. Apply unchanged transform

What happens and why: Centered query feature= $8-6=2 \text{ cm}^2$.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Centered query feature= $8-6=2 \text{ cm}^2$. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Area is a geometric toy feature. Useful predictive relationships need separate evaluation, and real leaves need a suitable measurement definition.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → **Diagnose** → **Repair**

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



COMPARE • DIAGNOSE • REPAIR

Tempting claim: Creating many features automatically gives the model more useful information. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record · **Expected** · **Actual** · **Likely cause / next test**

Evidence record: normal case

Expected: passes

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: defined behavior

Actual: _____

Likely cause / next test: _____

Evidence record: missing case

Expected: safe fallback

Actual: _____

Likely cause / next test: _____

Evidence record: shifted case

Expected: measured limit

Actual: _____

Likely cause / next test: _____

CLAIM TO CHALLENGE Creating many features automatically gives the model more useful information.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

An identifier may appear predictive because related objects leak across splits. Inspect grouped records and proxy risks before adding it. Never include private student details for this exercise; use the supplied fictional measurements.

How would you test adding area without claiming it must help?

Apply the connected idea: Relationships and confounding

A fairer comparison could offer the same task, instructions and assessment and randomly assign fictional groups to practice conditions. In real classrooms, a teacher must approve any activity and protect equal learning opportunities; this lesson uses a paper plan.

Even a better comparison has only four fictional groups. What should be reported?

8. Practise together

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



FREEZE THE EXPERIMENT

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Build a feature log with name, units, source, time available, transformation and missing-value policy. If width is missing, follow a declared training-derived imputation or return a clear unavailable result. Do not silently use zero, which describes a zero-area object.

Apply the connected idea: Relationships and confounding

A chart title Practice causes success overstates the evidence. Replace it with Practice time and scores in three fictional records, and describe the possible confounder. Better wording does not repair the study, but it makes the report honest.

Practise with less help: Transform validation

Use workbook W2 for the connected example “Feature engineering”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Feature engineering. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Relationships and confounding

Use its named workbook W2. Identify the given inputs, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Explain the picture

What should be saved so another group transforms the query identically? Point to the relevant step in the

picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Feature Workshop: design two transformations, compute examples, audit leakage, and compare one controlled validation result.

- 1. Write the hypothesis and expected evidence before running.
- 2. Feature Workshop: design two transformations, compute examples, audit leakage, and compare one controlled validation result.
- 3. Run one normal, one boundary, one missing, and one shifted case.
- 4. Calculate the metric from raw counts or row-level errors.
- 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

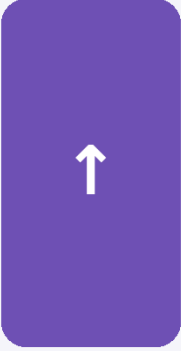
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



ML BUILDER MASTERY

1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

DEFINE Use all four terms precisely: transformation, derived feature, aspect ratio, feature pipeline.

CALCULATE Reproduce the worked evidence with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Creating many features automatically gives the model more useful information.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

11. Connect your project

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.



RESPONSIBLE PLANT CLASSIFIER

Add only justified Plant Classifier features, with formula, source fields, expected benefit, and leakage test.

THIS WEEK'S CHECKPOINT Add only justified Plant Classifier features, with formula, source fields, expected benefit, and leakage test.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

Evidence for your project

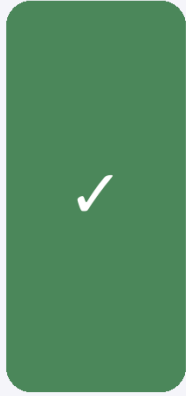
For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



INDEPENDENT ML BUILDER PROOF

Create and justify two plant measurement features.

REQUIRED PROOF Create and justify two plant measurement features.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Improve Generalization.

13. Study a complete case

Week 29 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Features are leaf length and width in cm; add length/width.

The question

Compute for 8 and 2; state boundary checks.

Worked explanation

Ratio=4, unitless with matching units. Guard zero/missing width; fit learned transforms on training only.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Engineer a ratio with boundary checks

Evidence or stage · Value or result

Evidence or stage: Length

Value or result: 8 cm

Evidence or stage: Width

Value or result: 2 cm

Evidence or stage: Length divided by width

Value or result: 4, unitless

Evidence or stage: Width zero or missing

Value or result: Do not divide, handle explicitly

Features are leaf length and width in cm; add length/width.

Compute for 8 and 2; state boundary checks.

Ratio=4, unitless with matching units. Guard zero/missing width; fit learned transforms on training only.

14. Try a new case independently

Independent case: Apply independently

Training areas [4, 8, 12]. A new rectangle has length 3 width 4. Compute the training mean and centered new feature; state units.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 29 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A feature is leaf_length divided by leaf_width. A row has width zero. What should the preprocessing rule do before division?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Add only justified Plant Classifier features, with formula, source fields, expected benefit, and leakage test.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Length=12 cm, width=30 mm. Compute length/width using matching units. Explain handling for zero or missing width and where learned scaling is fitted.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.